

Optimization of a UAV for Wildfire Management

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Dissertation submitted to the Faculty of the
Virginia Polytechnic Institute and State University
in partial fulfillment of the requirements for the degree of

Master of Science
in
Mechanical Engineering

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December XYZ, 2025

Blacksburg, Virginia

Keywords: Some Keywords, Subject matter, etc.

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(ABSTRACT)

Give a brief description of your thesis here.

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(GENERAL AUDIENCE ABSTRACT)

You are also required as of Spring 2016 to include a general audience abstract. This should be geared towards individuals outside of your field that may be reading seeking information about your work. You should avoid language that is particular to your field and clearly define any terms that may have special meaning in your discipline.

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NLP is a field of computer science, artificial intelligence, and linguistics concerned with the interactions between computers and human (natural) languages.

σ is the eighteenth letter of the Greek alphabet, and carries the 's' sound. In the system of Greek numerals, it has a value of 200.

Chapter 1

Introduction

1.1 One Section

1.1.1 A sub-section

1.2 Another Section

Chapter 2

Review of Literature

Chapter 3

UAV Design

3.1 UAV Geometry

The UAV is modeled as a simplified octocopter frame. [Talk about original UAV design??? [reference???]] The UAV geometry is shown in figure 3.1 - figure 3.3 with the UAV frame components highlighted. The frame model contains 3 components: the arms, the struts, and the hubs. The arms and struts are both beams with annular cross sections. The hub is two octagonal plates connected with a web shown in figure ???. The hub model is an simplification, as it can not fit the entire battery within the bounds of hub. Further work can be done to create a more detailed model.

The hub has 3 design variables, the web thickness, the circumscribed hub radius, and the thickness of the plates. Figure [ref] shows the design variables is [figure showing DVs] The hub height is a parameter set to 64mm as to fit the with UAV battery. [citation needed]

The arm beams span from the faces of hub to the wingspan. The design variables are thickness and radius. The wingspan is 1127.3mm. It is the minimum length required to fit all 8 propellers with a radius of [440mm??] and a propeller spacing of 25mm.

The strut beams have 3 design variables, thickness, radius, and, the distance from the hub center to the arm end. Strut deign variables are shown in figure [XYZ fig].

The UAV carries a payload of 45kg. All frame components are made from carbon fiber with

material properties shown in table [ref XYZ]. [DO I CITE EMAIL? CITATION NEEDED]

From these design variables relevant cross-section properties are calculated.

The drone batteries are also have a mass of [mass] and dimensions of [dimension]

il The DroneGeometry class within the UAV_Geometry module stores relevant design variables for a drone geometry. The relevant coordinates for hub, arm, and struts are calculated.

The projected surface area for drag calculation a drone geometry is calculated using the calc_projected_surfac_area method. [talk about this in detail later]

3.2 Mission Profile

A model of the mission profile was created to describe a potential mission with continuous position, velocity, and acceleration with respect to time. The mission profile is modeled as a series of mission profile segments. Constraints are put on the position, velocity, and acceleration at the nodes between mission profile segments. The remaining values at the nodes are calculated using basic kinematic equations. At values of time between mission segment nodes, the position, velocity, and acceleration are interpolated using cubic Hermite splines. [citation needed] Acceleration changes linearly with respect to time.

The mission profile has an initial takeoff period, a climb to an 100 meter altitude, a cruise period where the UAV travels 2000 meters downrange and increases in altitude 10 meters. Then the UAV descends for 20 seconds and then lands. The climb, cruise, and descent periods are each described using two separate cubic Hermite spline intervals to prevent over-defining the mission profile. Figure 3.5 shows the altitude for the mission profile against the down range position. Figure ?? shows the altitude over time. Figure ?? shows the downrange position with respect to time. Table 3.1 shows the constraints on the mission

profile. The duration shows the time between spline intervals. Blank entries are where no constraint was applied.

Segment	Initial	Takeoff	Climb 1	Climb 2	Cruise 1	Cruise 2	Decent 1	Decent 2	Land
Duration (s)	N/A	12	10	10	25	25	10	10	15
Position X (m)	0			550	1250	2000			2400
Position Y (m)	0			100	105	110			0
Velocity X (m/s)	0								0
Velocity Y (m/s)	0								0
Acceleration X (m/sec^2)	0			0		0			0
Acceleration Y (m/s^2)	0			0		0			0

Table 3.1: Mission Profile Constraints

Mission profiles are generated with a `mission_profile` module. This module contains a `MissionProfile` class, `MissionProfileSegment` class, and the kinematic matrix function. The `MissionProfile` contains a list of constraints added by the `add_constraint` method, and a list of segment durations added by the `add_segment` method. It also contains the `solve_kinematics` and `time_solve` method.

The `solve_kinematics` method assembles matrixes generated from the `kinematic_matrix` function into a kinematic matrix for the entire mission profile. The remaining position, velocity, and acceleration values at the nodes are then calculated using the `numpy.linalg.lstsq` function.

The `time_solve` method uses a cubic hermite spline to find interpolate values at a given time.

3.3 UAV Batteries

- jut put the

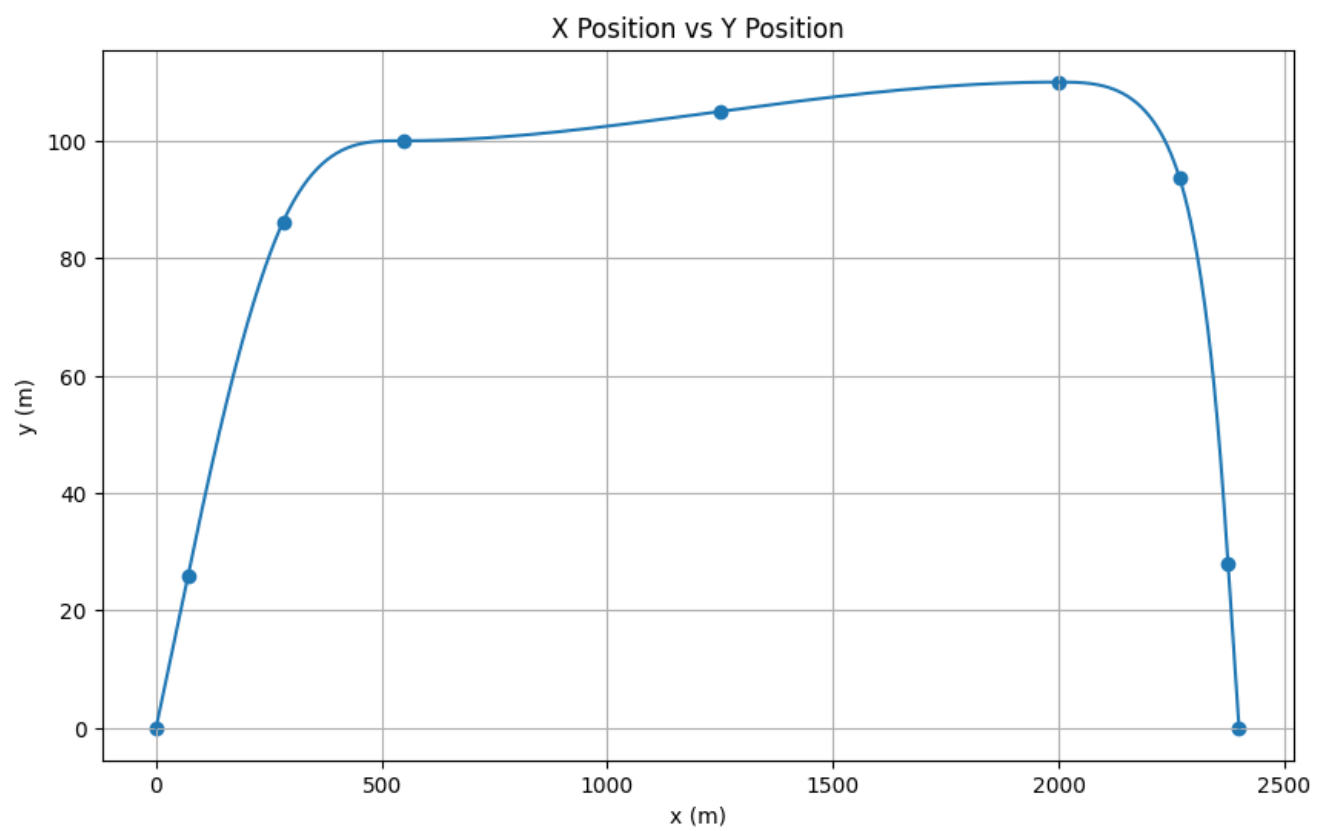


Figure 3.5: Downrange Position vs. Altitude

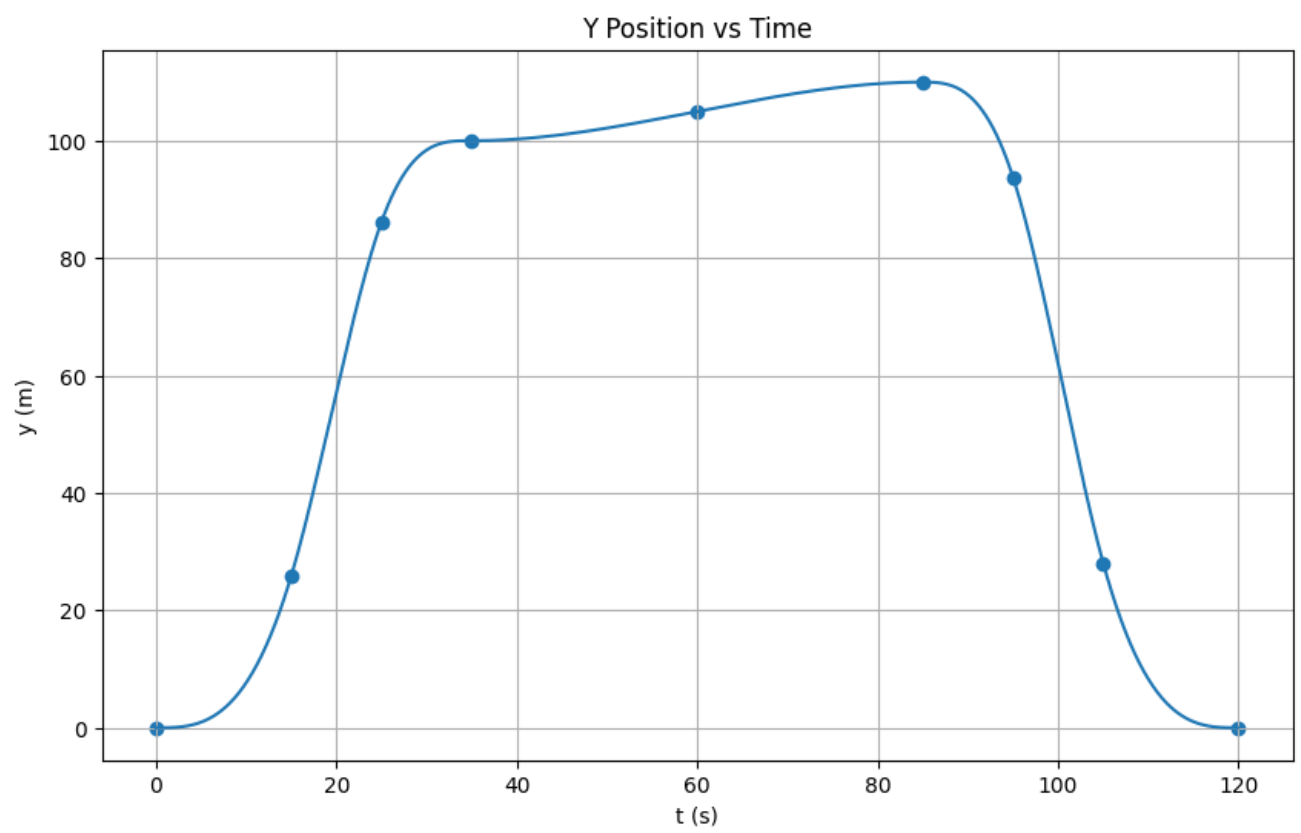


Figure 3.6: Altitude vs. Time

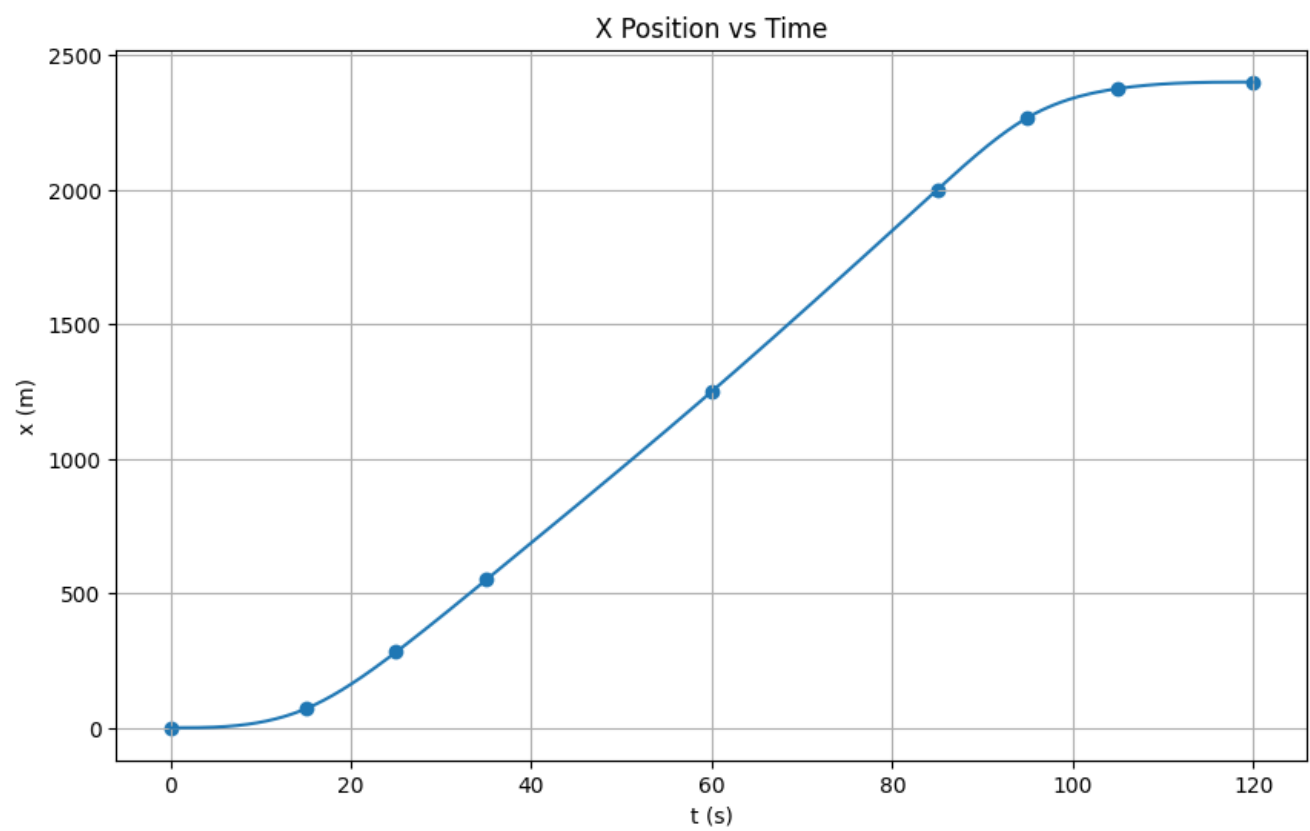


Figure 3.7: Downrange position vs. Time

Chapter 4

Finite Element Analysis

A finite element model was create to be run repeatedly by the optimization algorithm.

4.1 Beam Element Formulation

Beams are modeled as timeshenko beams elements with 12 degrees of freedom [citation needed?]. This beam model takes shear deformation effects into account. The mass for each beam is modeled as a 3D lumped mass matrix. Boundary conditions are set relative to a local coordinate system so that the symmetric boundary conditions at the endpoints of the struts.

4.2 Solvers

An FEA model was crated with three separate solvers. The static solver, which is used to calculate the response given a static loading, the dynamic solver, which calculate thee response given a force of an oscillating force at a given frequency, and the transient solver which calculates the response given a time-changing force.

4.2.1 Static

The static solver uses point loads to - Inputs - Point Loads - Outputs - Stress - Explain how stress calculation is obtained - calculated from the local displacement in each element - Strain - Explain how strain calculation is obtained - calculated from the local displacement in each element - Natural Frequency

4.2.2 Dynamic

- Inputs - Point Loads - Frequency [Cite source] - Outputs - Stress Amplitude - calculated from the local displacement in each element - Strain Amplitude - calculated from the local displacement in each element - Natural Frequency

4.2.3 Transient

- Inputs Function describing a force at a given time - Uses Newmark integration to calculate resulting transient displacements and reactions - The transient stress is obtained from the transient displacements and reactions

4.3 UAV Segment Model

A FEA model of single slice of the UAV frame was created. There are symmetric constraints at the midpoints of the strut beams. The center of the UAV is fixed in all 6 degrees of freedom. This assumes that the modes of frequency are symmetric. [maybe write about an asymmetric frequency model if I do that analysis] The arm and struts have an annular cross section. The hub is represented as a series of repeated I-beam cross sections. The width of

each I-beam cross-section in the FEA model is the width at the centroid of the corresponding trapezoidal area of the hub.

The UAV segment model is generated in the DroneFEA class within the drone_fea module. This module requires a DroneGeometry object as input. A BeamSystem object is generated to perform the FEA analysis. The nodes are generated using the create_drone_slice_node method which adds nodes to the BeamSystem object. The create_drone_slice_beams method calculates the hub section parameters and adds the beams to the BeamSystem object. The boundary_conditions_slice method applies the boundary conditions to the beam model.

4.4 Optimization

The problem is formulated as follows:

$$\min \text{mass}(d_a, t_a, d_s, t_s, r_s, t_f, t_w)$$

subject to:

$$g_j(x) \leq 0; j = 0, 1, 2, \dots, J$$

$$x_i^L \leq x_i \leq x_i^U$$

design variables:

- d_a : arm diameter,
- t_a : arm thickness,

- d_s : strut diameter,
- t_s : strut thickness,
- r_s : strut distance from center,
- t_f : hub flange thickness,
- t_w : hub web thickness
- h_l : hub length

derived variables:

- d_{tip} : Deflection at tip
- σ_{max} : Maximum stress from bending
- f_{amp} : Amplitude of displacement at the tip of the uav arm
- D : Damage per mission
- E_{used} : Energy used over th duration of the mission
- E_{total} : Energy used over th duration of the mission

design parameters:

- $d_{allowable}$: Deflection allowable
- $\sigma_{allowable}$: Stress allowable from bending
- $R_{allowable}$: Allowable amplitude displacement ratio
- $n_{missions}$: number of missions before damage

Constraints:

Limit on deflection from bending by force required to lift the UAV.

$$g_1(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = d_{tip} \leq d_{allowable}$$

Limit on stress from bending by force required to lift the UAV.

$$g_2(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = \sigma_{max} \leq \sigma_{allowable}$$

Limit on amplitude of displacement given a forcing function based on the propeller speed of the UAV.

$$g_3(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = \frac{f_{amp}}{d_{tip}} \leq R_{allowable}$$

Limit on the damage the UAV during a mission.

$$g_4(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = D * n_{missions} \leq 1$$

Energy

$$g_5(d_a, t_a, d_s, t_s, r_s, t_f, t_w, h_l) = E_{used}/E_{total} \leq 0.8$$

The bounds on design variable were determined by limits on commercially available carbon fiber based on the dragon plate website.

$$9\text{mm} \leq r_a \leq 50.0\text{mm}$$

$$0.254\text{mm} \leq t_a \leq 3.175\text{mm}$$

$$9\text{mm} \leq r_s \leq 50.0\text{mm}$$

$$0.254\text{mm} \leq t_s \leq 3.175\text{mm}$$

$$400\text{mm} \leq d_s \leq 1127.3\text{mm}$$

$$100\text{mm} \leq h_l \leq 1127.3\text{mm}$$

- Optimization algorithm used is SLSQP method using scipy's minimization function.

4.4.1 Objective Function and Design Variables

4.4.2 Constraints

Displacement

- 1/8 of the mass of the frame and payload was modelled as a point load at the end of the UAV arm pointing upward while the center of the UAV

is fixed - The static solver as used to calculate the maximum displacement magnitude.

$$g_1(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = d_{tip} \leq d_{allowable}$$

Stress

- The stress calculation is the same method as the displacement calculation.
- The allowable stress is determined by the maximum stress of the material and a factor of safety.

$$g_2(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \sigma_{max} \leq \sigma_{allowable}$$

Frequency

- The goal of the frequency constraint is to make sure that the frequency experienced by the propeller is not near the natural frequency of the vehicle
- The dynamic solver is used. - A frequency of 2 * the rad/s of the propeller is used as the propeller has 2 blades. - A force of 1 Neewton is input at the tip of the UAV bladd [frequency plot]

$$g_3(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = \frac{f_{amp}}{d_{tip}} \leq R_{allowable}$$

Damage

- Mission profile with 5s rev up rev down - Oscillating force based on propeller frequency - [figure of oscillating stress] - Raincounting

$$g_4(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = D * n_{missions} \leq 1$$

Energy

- Mission profile - Power to trust calculation - Battery capacity

$$g_5(d_a, t_a, d_s, t_s, r_s, t_f, t_w) = E_{used}/E_{total} \leq 0.8$$

Chapter 5

Results

- The initial guess were based off of measurements from the original UAV design. - arm diameter, strut diameter, hub radius, flange thickness, and web thickness were all found to converge to boundary conditions
- Sensitivity analysis - The value of the design variables which did not converge to the boundary conditions were varied by $\pm 0.1\%$ of thier values to test the optimal solution.

Table 5.1

Chapter 6

Conclusions

Chapter 7

Summary

Appendices

Appendix A

First Appendix

A.1 Section one

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A.2 Section two

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Appendix B

Second Appendix

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